VELOOP REWARDS - Product Data Audit

Data Analyst Internship, Task 01 | Work Package A

Authenticity notice: This audit maps the data VELOOP *will eventually need*. Every event, field and feature below is **Proposed** or **To Be Verified**. Nothing is claimed to be implemented, and no simulated or real VELOOP statistics appear in this report.

1. Purpose and method

The goal is a complete map of what VELOOP must measure so that analysis is possible once real activity exists. The task brief lists candidate product areas (accounts, tasks, earning features, daily engagement, referrals, wallet, withdrawals, support) but states that not every feature may be implemented. I therefore used the brief as the only source, treated all features as unverified, and recorded open questions in the Assumption Register (Step 1, A001-A018). An attempt to find official public documentation for VELOOP Rewards returned no matching official page (see Step 1, S008).

2. Summary of findings

- **20 product areas** mapped, from Registration to third-party earning providers (Section 3).
- **29 user-action steps** across 9 features, with a decision on which should become measurable events (Section 4).
- **User event, system event and business outcome** separated for 21 items. The key rule: a user finishing an activity is not the same as a reward being credited, and registration is not the same as referral conversion (Section 5).
- **42-row Data Coverage Matrix** aligned with the Event Tracking Plan, each with a proposed event, source, business question and priority; 9 rows rated Critical (Section 6, and the Excel file 03_Data_Coverage_Matrix).
- **15 analytics gaps** that could affect future production reporting: 5 Critical, 7 High, 2 Medium, 1 Low (Section 7).

3. Product area mapping (A1)

ID	Product Area	Feature	User Action	System Outcome	Data Required	Business Question	Status
PA01	Account	Registration	User signs up	Account created; user_id issued	user_id, registered_at, registration_source, platform, referral_code_used	How many users sign up and through which source?	To Be Verified
PA02	Account	Login	User logs in / out	Session started or ended	user_id, login_at, platform, session_id (proposed)	How often do registered users come back?	To Be Verified
PA03	Account	Onboarding	User starts / completes onboarding steps	Onboarding state updated	user_id, onboarding_step, step_started_at, step_completed_at	Where do new users drop off before first earning?	Proposed
PA04	Account	Account status	(none; system or admin action)	Status changed (e.g. active, restricted, closed)	user_id, old_status, new_status, changed_at, reason_category	How many accounts change state and why?	To Be Verified
PA05	Tasks	Task catalogue	View, open, start, complete a task	Completion validated; reward credited or rejected	task_id, task_type, user_id, start/complete time, validation_status, reward_amount, reward_unit	Which tasks get started, finished and rewarded?	To Be Verified
PA06	Earning	Watch Ads	Starts and finishes an ad view	Ad provider returns completion signal; reward credited	ad_provider, ad_placement_id, ad_started_at, ad_completed_flag, reward_amount	What share of ad views end in a credited reward?	To Be Verified
PA07	Earning	Tap & Earn	Taps / collects	Tap count or energy recorded; reward credited	user_id, tap_session_id, tap_count, reward_amount, reward_unit	How much of the user base uses tap-based earning, and how often?	To Be Verified

ID	Product Area	Feature	User Action	System Outcome	Data Required	Business Question	Status
PA08	Earning	Mining	Starts a cycle; claims result	Cycle completes; reward credited on claim	mining_cycle_id, start_time, end_time, claimed_at, reward_amount	Do users return to claim and restart cycles?	To Be Verified
PA09	Engagement	Daily Rewards	Claims daily reward	Reward credited; streak updated	user_id, claim_date, streak_day, reward_amount	Do daily rewards build repeat usage?	To Be Verified
PA10	Engagement	Streaks	(follows daily activity)	Streak incremented or reset	user_id, streak_count, previous_streak_count, change_reason	How long do streaks last and when do they break?	To Be Verified
PA11	Engagement	Spin	Starts a spin	Outcome determined; reward credited	spin_id, spin_started_at, outcome_category, reward_amount, reward_unit	How often is Spin used and what is the reward distribution?	To Be Verified
PA12	Engagement	Missions	Starts / progresses a mission	Mission completed; reward credited	mission_id, mission_type, progress_value, completed_at, reward_amount	Which missions are completed vs abandoned?	To Be Verified
PA13	Engagement	Badges / Levels / XP	(follows activity)	Badge earned; XP added; level changed	user_id, badge_id, xp_amount, xp_source, level_before, level_after	Does progression drive further activity?	To Be Verified
PA14	Currency	Reward units (VEs, Gems, Tokens, XP, Spins)	(none)	Balances change	reward_unit, unit_type, balance_after	Which units exist, what are they for, can they convert?	To Be Verified
PA15	Referrals	Referral programme	Shares code; friend registers	Referral registered, activated, converted	referral_id, referrer_user_id, referred_user_id, created_at, registered_at, activated_at, status	How many referrals become genuinely active users?	To Be Verified
PA16	Wallet	Wallet transactions	Views wallet	Transaction recorded for every balance change	transaction_id, user_id, type, amount, unit, status, reference_id, related_transaction_id	Do wallet balances reconcile with rewards and withdrawals?	To Be Verified
PA17	Wallet	Exchange	Exchanges one unit for another	Paired debit and credit recorded	from_unit, to_unit, from_amount, to_amount, exchange_rate_applied	How much exchange activity happens and at what rates?	To Be Verified
PA18	Withdrawals	Withdrawal request	Requests withdrawal	Review, then completed / rejected / reversed	withdrawal_id, requested_amount, method, requested_at, processing_at, final_at, status, reason_category	How many requests complete, and how long do they take?	To Be Verified
PA19	Support	Help Center / support requests	Views article; submits request	Ticket created and resolved	ticket_id, issue_category, created_at, resolved_at, article_id	What do users struggle with?	To Be Verified
PA20	Third parties	Ad / survey / offer / game providers	(indirect)	Provider returns success, failure or delay	provider_name, provider_reference_id, provider_status, callback_received_at	How much data loss or delay comes from external providers?	To Be Verified

4. User action sequences (A2)

For each feature, the steps the user or system goes through, and whether the step should become a measurable event. Steps marked 'No' are merged into a neighbouring event to avoid counting unvalidated activity as success.

Feature	Step	Step Name	Actor	Measurable event?	Proposed Event	Reason
Tasks	1	Task viewed	User	Optional	task_viewed	Needed for discovery-to-start conversion; high volume, so sample if cost matters
Tasks	2	Task opened	User	Yes	task_opened	Shows intent beyond a view
Tasks	3	Task started	User	Yes	task_started	Denominator for completion rate
Tasks	4	Task completed (user side)	User	No (merge into step 5)	-	Completion before validation must not be counted as success
Tasks	5	Completion validated	System	Yes	task_completed	Fire only after validation passes (as brief example)
Tasks	6	Validation failed / user abandoned	System / User	Yes	task_failed	Needed to explain the gap between starts and completions
Tasks	7	Reward credited	System	Yes	reward_credited	Separate from completion: activity occurred is not reward credited
Watch Ads	1	Ad requested / shown	System	Optional	ad_shown	Provider may fail to serve; measures fill problems
Watch Ads	2	Ad completed	User + provider	Yes	ad_completed	Depends on provider signal; record source
Watch Ads	3	Reward credited	System	Yes	reward_credited	Links to wallet transaction
Daily Rewards	1	Daily reward claimed	User	Yes	daily_reward_claimed	Core daily activity signal
Daily Rewards	2	Streak updated	System	Yes	streak_updated	Records both increments and resets
Spin	1	Spin started	User	Yes	spin_started	Adoption measure
Spin	2	Spin completed	System	Yes	spin_completed	Outcome and reward
Mining	1	Cycle started	User	Yes	mining_started	Start of repeat-usage loop
Mining	2	Reward claimed	User	Yes	mining_claimed	Shows return behaviour
Missions	1	Mission started	User	Yes	mission_started	Denominator for mission completion
Missions	2	Mission completed	System	Yes	mission_completed	System decides completion
Registration	1	User registered	User + system	Yes	user_registered	Start of lifecycle
Registration	2	Onboarding started	User	Yes	onboarding_started	Onboarding funnel entry
Registration	3	Onboarding completed	User	Yes	onboarding_completed	Activation proxy (definition to be agreed)
Referrals	1	Referral created	User	Yes	referral_created	Code or link generated / shared
Referrals	2	Referred user registered	System	Yes	referral_registered	Registration is not conversion
Referrals	3	Referred user activated	System	Yes	referral_activated	Needs agreed definition of activated (A010)
Referrals	4	Referral converted	System	Yes	referral_converted	Business outcome of referral
Withdrawals	1	Requested	User	Yes	withdrawal_requested	Start of withdrawal lifecycle
Withdrawals	2	Under review / processing	System	Yes	withdrawal_processing	Needed for processing-time KPI

Feature	Step	Step Name	Actor	Measurable event?	Proposed Event	Reason
Withdrawals	3	Completed	System	Yes	withdrawal_completed	Business outcome
Withdrawals	4	Rejected / reversed	System	Yes	withdrawal_rejected / withdrawal_reversed	Alternative end states; each needs reason category

5. User event vs system event vs business outcome (A3)

Item	Classification	Why	Proposed Event
User opens a task	User Event	Initiated by the user	task_opened
User starts a task	User Event	Initiated by the user	task_started
User claims daily reward	User Event	Initiated by the user	daily_reward_claimed
User requests withdrawal	User Event	Initiated by the user	withdrawal_requested
User starts a spin	User Event	Initiated by the user	spin_started
User logs in	User Event	Initiated by the user	user_logged_in
Task completion validated	System Event	Decided by system rules	task_completed
Task validation failed	System Event	Decided by system rules	task_failed
Reward credited	System Event	Wallet balance changed by system	reward_credited
Reward pending	System Event	Held before crediting	reward_pending
Reward reversed	System Event	Credit undone by system or operations	reward_reversed
Withdrawal under review	System Event	Operations or automated check	withdrawal_processing
Withdrawal rejected	System Event	Decision by system or operations	withdrawal_rejected
Account status changed	System Event	Admin or system action	account_status_changed
Streak updated	System Event	Computed by system	streak_updated
Referral activated	System Event	System detects qualifying activity	referral_activated
Mission completed	System Event	System evaluates progress	mission_completed
Daily active users increase	Business Outcome	Derived metric, not an event	(KPI)
Withdrawal completed and paid out	Business Outcome	Final result of a lifecycle	withdrawal_completed
Referral converted	Business Outcome	Final result of a referral lifecycle	referral_converted
Task completion rate	Business Outcome	Derived: completed / started	(KPI)

6. Data Coverage Matrix (A4)

Event names, sources, statuses and priorities match the Event Tracking Plan (object_action naming, past tense). Names are proposals only.

Product Area	User Action	System Action	Business Outcome	Data Required	Proposed Event	Source	Business Question	Status	Priority
Account	Signs up	Creates account, issues user_id	New user	user_id, registered_at, registration_source, platform	user_registered	Backend	Volume and source of new users?	To Be Verified	Critical
Account	Logs in	Starts session	Returning activity	user_id, login_at, platform, session_id	user_logged_in	Backend	How often do users return?	Proposed	High
Account	Logs out	Ends session	-	user_id, logout_at	user_logged_out	Frontend	Needed only where session length matters	Proposed	Low
Account	Starts onboarding	Records step	Funnel entry	user_id, onboarding_step, started_at	onboarding_started	Frontend	Where do new users enter the funnel?	Proposed	High
Account	Completes onboarding	Marks complete	Activation proxy	user_id, completed_at	onboarding_completed	Backend	What share finishes onboarding?	Proposed	High
Tasks	Views a task	-	Discovery	user_id, task_id, task_type, feature	task_viewed	Frontend	What do users see vs start?	Proposed	Medium
Tasks	Opens a task	-	Intent	user_id, task_id, task_type	task_opened	Frontend	Which tasks attract attention?	Proposed	Medium
Tasks	Starts a task	Creates task activity record	Task start	user_id, task_id, task_type, started_at, task_activity_id	task_started	Backend	How many tasks begin?	Proposed	Critical
Tasks	Completes a task	Validates completion	Task completed	user_id, task_id, completed_at, status, reward_amount, reward_unit	task_completed	Backend	Completion rate per task type?	Proposed	Critical
Tasks	Abandons or fails	Records failure class	Task failed	user_id, task_id, failure_category	task_failed	Backend	Why do tasks not finish?	Proposed	High
Rewards	-	Credits reward to wallet	Reward credited	reward_id, user_id, amount, unit, source_feature, wallet_transaction_id	reward_credited	Backend	How much is credited, by feature?	Proposed	Critical
Rewards	-	Holds reward until validated	Reward pending	reward_id, pending_reason, expected_resolution_at	reward_pending	Backend	How many rewards stay pending and for how long?	Proposed	High
Rewards	-	Reverses a credited reward	Reward reversed	reward_id, original_reward_id, reason_category, reversed_at	reward_reversed	Backend	How often do credits get undone?	Proposed	Critical
Rewards	-	Corrects a reward amount	Reward corrected	reward_id, old_amount, new_amount, correction_reason	reward_corrected	Backend	Are rewards being corrected after the fact?	Proposed	Medium
Daily Rewards	Claims daily reward	Credits reward	Daily activity	user_id, claim_date, streak_day, reward_amount	daily_reward_claimed	Backend	Does the daily reward drive habit?	Proposed	High
Streaks	-	Updates streak	Streak change	user_id, streak_count, previous_streak_count, change_type	streak_updated	Backend	Where do streaks break?	Proposed	Medium
Spin	Starts spin	-	Adoption	user_id, spin_id	spin_started	Frontend	How many users spin?	Proposed	Medium
Spin	-	Determines outcome	Spin result	spin_id, outcome_category, reward_amount, reward_unit	spin_completed	Backend	What does Spin pay out?	Proposed	Medium
Missions	Starts mission	Records start	Mission start	user_id, mission_id, started_at	mission_started	Backend	Which missions are taken up?	Proposed	Medium
Missions	-	Marks mission complete	Mission completed	user_id, mission_id, completed_at, reward_amount	mission_completed	Backend	Which missions finish?	Proposed	Medium

Product Area	User Action	System Action	Business Outcome	Data Required	Proposed Event	Source	Business Question	Status	Priority
Badges / Levels	-	Awards badge	Progression	user_id, badge_id, earned_at	badge_earned	Backend	Does progression affect activity?	Proposed	Low
Referrals	Shares code	Creates referral record	Referral created	referral_id, referrer_user_id, created_at, source	referral_created	Backend	How many referrals are started?	Proposed	High
Referrals	-	Links new user to referral	Referral registered	referral_id, referred_user_id, registered_at	referral_registered	Backend	How many invitees register?	Proposed	High
Referrals	-	Detects qualifying activity	Referral activated	referral_id, activated_at, activation_rule	referral_activated	Backend	How many invitees become active?	To Be Verified	High
Referrals	-	Marks referral successful	Referral converted	referral_id, converted_at, reward_transaction_id	referral_converted	Backend	What is the true referral conversion?	Proposed	High
Wallet	-	Creates transaction	Balance change	transaction_id, user_id, type, amount, unit, status, reference_id	wallet_transaction_created	Backend	Does every balance change have a record?	To Be Verified	Critical
Wallet	-	Completes transaction	Balance final	transaction_id, completed_at	wallet_transaction_completed	Backend	How long until transactions settle?	To Be Verified	High
Wallet	-	Reverses transaction	Reversal	transaction_id, related_transaction_id, reason_category	wallet_transaction_reversed	Backend	How often are transactions reversed?	To Be Verified	Critical
Wallet	Exchanges units	Creates paired debit and credit	Exchange	from_unit, to_unit, from_amount, to_amount, rate_applied	wallet_exchange_completed	Backend	How much exchange activity exists?	To Be Verified	Medium
Withdrawals	Requests withdrawal	Creates request	Withdrawal request	withdrawal_id, user_id, requested_amount, method, requested_at	withdrawal_requested	Backend	How many users try to withdraw?	To Be Verified	Critical
Withdrawals	-	Moves to review	In review	withdrawal_id, processing_at	withdrawal_processing	Backend	How big is the pending queue?	To Be Verified	High
Withdrawals	-	Completes payout	Completed	withdrawal_id, final_at, status	withdrawal_completed	Backend	How many requests complete?	To Be Verified	Critical
Withdrawals	-	Rejects request	Rejected	withdrawal_id, final_at, reason_category	withdrawal_rejected	Backend	Why are requests rejected?	To Be Verified	High
Withdrawals	-	Reverses payout	Reversed	withdrawal_id, reversed_at, reason_category	withdrawal_reversed	Backend	How often are payouts reversed?	To Be Verified	High
Support	Opens Help Center / submits request	Creates ticket	Support demand	ticket_id, issue_category, created_at, article_id	support_request_created	Backend	What do users need help with?	To Be Verified	Low
Third-party earning	-	Receives provider callback	Provider outcome	provider_name, provider_reference_id, provider_status, callback_received_at	provider_callback_received	Provider	How reliable are external providers?	To Be Verified	High
Account	-	Status changed by user, system or operations	Account state change	user_id, previous_status, new_status, changed_at, status_change_actor	account_status_changed	Backend	How many accounts change state, and why?	To Be Verified	High
Rewards	-	Refuses a pending reward	Reward not credited	reward_id, reason_category	reward_rejected	Backend	How many pending rewards are never credited?	To Be Verified	Medium
Mining	Starts a mining cycle	Creates cycle record	Cycle start	mining_cycle_id, started_at	mining_started	Backend	Do users start cycles?	To Be Verified	Medium
Mining	Claims cycle result	Credits reward	Return behaviour	mining_cycle_id, reward_amount, reward_unit, claimed_at	mining_claimed	Backend	Do users return to claim?	To Be Verified	Medium

Product Area	User Action	System Action	Business Outcome	Data Required	Proposed Event	Source	Business Question	Status	Priority
Feature usage	Opens a feature screen	-	Feature adoption	feature_name, screen_name	feature_opened	Frontend	How many users adopt features that have no start event of their own?	To Be Verified	Medium
Withdrawals	Cancels request before processing	Cancels withdrawal	Cancelled	withdrawal_id, reason_category	withdrawal_cancelled	Backend	How many requests are cancelled by users?	To Be Verified	Low

7. Analytics gaps

Each priority carries a reason. Critical means reporting would be unreliable or impossible without it.

Gap ID	Analytics Gap	Area	Why It Could Affect Production Reporting	Priority	Priority Reason	Linked Assumption
G01	No confirmed stable user identifier	Account	Cross-event, retention and cohort analysis break if user_id changes or is duplicated across devices	Critical	Without a stable user_id, one person can appear as several users	A003
G02	No confirmed unique transaction ID on wallet entries	Wallet	Duplicate credits cannot be detected or reconciled	Critical	Without a stable transaction ID, reliable wallet deduplication cannot be guaranteed	A005
G03	Unknown whether task completion events are emitted	Tasks	Task KPIs cannot be computed from production data	Critical	All earning analytics depend on completion events	A001
G04	Reward unit definitions are undefined (VE, XP, Gems, Tokens, Spins)	Currency	Amounts in different units cannot be summed or compared	Critical	Wrong totals would mislead management	A002
G05	Activity vs credited reward not clearly separated	Rewards	Reward KPIs may count activity as paid rewards	Critical	Overstated reward figures	A013
G06	Withdrawal state history may not be stored	Withdrawals	Processing time and rejection analysis need timestamps for every state	High	Without state history, only the latest status is visible	A006, A007
G07	No agreed definition of an activated or converted referral	Referrals	Registration could be reported as conversion	High	Inflated referral success	A010
G08	Session identifier not confirmed	Account	Session-level analysis not possible; DAU still possible via events	Medium	Useful but not blocking for core KPIs	A004
G09	Failure reasons for tasks and withdrawals not classified	Tasks / Withdrawals	Cannot explain drop-offs or rejections	High	Needed to act on problems, not just count them	A001, A007
G10	Third-party provider data (ads, surveys, offers) may be partial or delayed	Third parties	Completions may be missing or late, understating activity	High	External systems are outside VELOOP control	A009
G11	Timestamp format and timezone not confirmed	All	Daily metrics may shift across day boundaries	High	Wrong day bucket changes DAU and streaks	A012
G12	Registration and acquisition source not confirmed	Account	Acquisition dashboards impossible	Medium	Valuable for marketing but not blocking	A014
G13	Feature live vs planned status unknown	All	Events may be built for features that do not exist	High	Wasted engineering effort and empty dashboards	A008
G14	No confirmed data store or event pipeline	Platform	Nowhere to land events for analysis	High	Pipeline design depends on it	A011
G15	Support / Help Center data may not exist	Support	Support KPIs may not be reportable	Low	Not needed for first release	A015

8. Limitations and next steps

- Source is the task brief only; no official VELOOP documentation or engineering confirmation has been seen.
- The simulated dataset has since been profiled and audited (Deliverables 12 and 13). It covers the Users entity only, so the other entities in this audit have not been tested against data.
- Next: Work Package B - User and Account Analytics Specification (identifiers, user fields, lifecycle).